SIMULATOR CONSOLE · CANDIDATE PROFILE

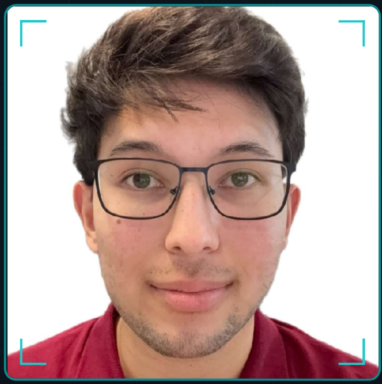
TRACKING ACTIVE X 11.53 · Y 48.10

CANDIDATE ID · AZ-2004

Assain Zhakypbek

Working Student — Machine Learning & Software Engineering

Computer Science student at TU Munich with hands-on **ML engineering** (LLM fine-tuning in the PyTorch ecosystem) and low-level **C++ / SystemC systems** work built from the ground up. Eager to help engineer Medability's instrument-**tracking system** and learn the medical imaging behind surgical simulation.



SCAN · 680PX ⊙ LOCKED

BORN	ORIGIN	NATIONALITY	BASED
12.03.2004	Astana, KZ	Kazakh	Munich, DE

01 CONTACT

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02 LANGUAGE CALIBRATION

- Kazakh — NATIVE
- Russian — NATIVE
- English — FLUENT
- German — C1

03 OFF-DUTY

- Gym
- Football · amateur league

04 PROCEDURE LOG · EXPERIENCE

Machine Learning Engineering Project — NOV 2025 – FEB 2026

Technical University of Munich · Munich, DE

88.4% PASS@1 · HUMANEVAL (+26.2%)

- Fine-tuned **Llama-3-8B** and Codegen-350M for Python function generation using **QLoRA** and **DoRA** via the Unsloth framework, engineering around local VRAM constraints.
- Curated and formatted training data into strict chat templates, driving the Pass@1 gain on the OpenAI HumanEval benchmark.
- Built a custom evaluation pipeline measuring functional correctness, AST-based syntax validity, and metamorphic consistency across prompt variations.
- Deployed a **GGUF**-quantized model via a **FastAPI** server with Apple Metal (MPS) acceleration, integrated with a custom IntelliJ IDEA plugin for real-time inference.

PyTorch · Python · Unsloth · QLoRA / DoRA · FastAPI · HumanEval

05 TRAINING MODULES · PROJECTS

Team Health & Analytics Platform

Management dashboard · team-health metrics

APR 2026 — PRESENT

- Architected the dashboard with **Feature-Sliced Design**, organizing the codebase into domain-driven layers to support complex team-health metrics.
- Used **Next.js App Router** and React Server Components for a high-performance frontend with optimized initial load.
- Built an API-ready architecture with modular state logic, ready to integrate live health-tracking data.

TypeScript · Next.js · React · Tailwind · Vercel

Cache Architecture Simulator

Cycle-accurate hardware-level simulator

JUN — JUL 2024

- Engineered a simulator for direct-mapped and 4-way set-associative caches, modeling clock-accurate write-through, FIFO replacement, and primitive gate-count estimation.
- Built a C CLI with **getopt_long** to configure simulation parameters and parse memory access traces.
- Implemented cross-language integration between a C frontend and **C++ / SystemC** backend via `extern "C"` bindings.

C++ · C · SystemC

06 PROFICIENCY MATRIX · SKILLS

PROGRAMMING

Python · C++ · Java · C · TypeScript · JavaScript · HTML / CSS

FRAMEWORKS & LIBRARIES

PyTorch · React · Next.js · Node.js · FastAPI · pandas · NumPy

TOOLS & PLATFORMS

Git · Docker · REST · VS Code · PyCharm · IntelliJ

highlighted = mapped to the Medability role brief

07 EDUCATION

Technical University of Munich

OCT 2023 — MAR 2027 (exp.)

B.Sc. Computer Science · Munich, Germany

RELEVANT COURSEWORK

Advanced Topics of Software Testing · Introduction to Emerging Computing Technologies